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CS379

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1. Refcount Trace:

d1 refcount = 1, d2 refcount = 0	`let a = Dog();`
d1 refcount = 2, d2 refcount = 0	`let b = a;`
d1 refcount = 2, d2 refcount = 1	`let c = Dog();`
d1 refcount = 1, d2 refcount = 2	`b = c;`
d1 refcount = 0, d2 refcount = 2	`a = null;`

D1 refcount is 0 whenever a is set to null, the memory must be freed as there is nothing in it anymore.

2. The cycle won't die:

n1 refcount = 1, n2 refcount = 0	`let n1 = Node();`
n1 refcount = 1, n2 refcount = 1	`let n2 = Node();`
n1 refcount = 2, n2 refcount = 1	`n1.next = n2;`

n1 refcount = 2, n2 refcount = 2	`n2.prev = n1;`
n1 refcount = 1, n2 refcount = 2	`n1 = null;`
n1 refcount = 1, n2 refcount = 1	`n2 = null;`

Since the references still exist to each other respectively, it will continue to point to the null object, meaning the refcount for both will always be at least 1

3. Refcounting vs Mark-and-Sweep:

Refcounting may be useful when opening and and reading files it can destroy the file handle and thus letting the file be open to other operations should the refcount = 0. Mark and Sweep may be useful for operations that contain circular dependency variables as it does not care about the refcount and thus is able to reach variables that point to the latter variable.